

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

SaaS Business Management System for WordPress

CSE 4104 — Software Development III

June 2026

1. Introduction

1.1 Project Overview

The SaaS Business Management Plugin is an AI-powered WordPress plugin designed to bring intelligent business operations directly into the wp-admin dashboard. It integrates inventory tracking, financial monitoring, and LLM-driven analytics into a unified interface for small and medium-sized businesses (SMBs).

1.2 Objectives

- Provide a centralized AI-powered business dashboard within WordPress.
- Automate inventory and stock management with demand forecasting.
- Consolidate sales and expense data for real-time financial visibility.
- Integrate an LLM API (GPT-4o / Gemini) for natural-language insights.
- Enable conversational querying of business data via an AI chat interface.

1.3 Scope

The system targets WordPress-based eCommerce businesses using WooCommerce. It operates entirely within the WordPress admin panel and communicates with external AI APIs over HTTPS. Out of scope: mobile apps, non-WordPress platforms, and direct payment processing.

2. Functional Requirements

FR-01: Authentication & Access Control

- The system shall enforce WordPress role-based access (Administrator, Manager).
- All sensitive pages shall be restricted to authorized roles only.
- The system shall maintain an activity log for all data access and modifications.

FR-02: Analytics Dashboard

- The system shall display real-time KPI cards: revenue, profit margin, top products, customer acquisition rate.
- The system shall render interactive charts comparing current vs. historical performance.

FR-03: Inventory & Stock Management

- The system shall display a centralized product inventory table with live stock levels.
- The system shall send automated low-stock alerts via admin notices and email.
- The system shall use AI demand forecasting to suggest reorder quantities and timing.

FR-04: Expense & Sales Tracking

- The system shall allow manual and automated expense entry with categories.
- The system shall aggregate WooCommerce sales data and compute profit/loss.
- The system shall export financial reports as CSV and PDF.

FR-05: AI Insights Engine

- The system shall auto-generate weekly and monthly performance summaries via LLM.
- The system shall surface smart recommendations (underperforming products, pricing, cost reduction).
- The system shall detect and alert on anomalies in revenue or spending patterns.

FR-06: AI Chat Assistant

- The system shall provide an embedded chat interface for natural-language data queries.
- Responses shall be grounded in the user's actual business data.
- Example queries: "What were my top 5 products last month?", "Which expense category grew most this quarter?"

FR-07: Notification System

- The system shall maintain an in-dashboard notification center for alerts and AI insights.
- The system shall optionally send email digests summarizing key metrics and action items.

3. Non-Functional Requirements

Category	Requirement
Performance	AI API responses shall complete within 5 seconds under normal load.
Security	All API keys stored encrypted; HTTPS enforced for all external calls.
Usability	Dashboard UI shall be fully responsive and accessible on tablet and desktop.
Reliability	Plugin shall not disrupt core WordPress functions if AI API is unavailable.
Scalability	Database schema shall support up to 10,000 product SKUs and 5 years of sales history.
Compatibility	Compatible with WordPress 6.x, WooCommerce 8.x, PHP 8.x.
Maintainability	Code shall follow WordPress Coding Standards and be documented with inline PHPDoc.

4. User Roles

Role	Permissions
Administrator	Full access: configure settings, manage all modules, view all reports, access AI chat.
Manager	Access to dashboard, inventory, sales tracking, and AI insights. Cannot change plugin settings.
Viewer (optional)	Read-only access to dashboard and reports. No edit or configuration rights.

5. Use Cases

UC-01: View Business Dashboard

Actor: Administrator / Manager

Precondition: User is logged into WordPress with appropriate role.

Main Flow: User navigates to plugin dashboard → System fetches KPIs from database → System renders charts and metric cards → User views real-time business overview.

Postcondition: Dashboard is displayed with up-to-date data.

UC-02: Receive Low-Stock Alert

Actor: System (automated)

Precondition: A product's stock level falls below the configured threshold.

Main Flow: System detects low stock → Triggers admin notice in WordPress dashboard → Sends email alert to Administrator → Logs event in notification center.

Postcondition: Administrator is informed and can initiate reorder.

UC-03: Generate AI Business Insights

Actor: System (scheduled) / Administrator

Precondition: Sufficient sales and expense data exists for the reporting period.

Main Flow: System collects period data → Sends structured prompt to LLM API → Receives natural-language summary → Stores and displays insight in dashboard notification center.

Postcondition: Administrator receives actionable weekly/monthly insight.

UC-04: Query Data via AI Chat

Actor: Administrator / Manager

Precondition: User is on the plugin page; AI API is available.

Main Flow: User types query (e.g., "Show top 5 products last month") → System formats query with business context → Sends to LLM API → Displays response in chat interface.

Postcondition: User receives a context-aware, data-grounded answer.

UC-05: Export Financial Report

Actor: Administrator / Manager

Precondition: Sales and expense data exists for the selected period.

Main Flow: User selects date range and format (CSV or PDF) → System compiles profit/loss data → Generates downloadable file → User downloads report.

Postcondition: Report file is downloaded to user's device.

